

Report

General work plan for both phases:-

1. Check the data to see amount of outliers, and to check if any degree is more prominent than others
2. Check the degree with the least MSE and std without any regularization.
3. Check best alpha for the degrees around the degrees found previously.
4. Use the model with the least errors to make the predictions

All files are uploaded to github.

Check figures/ directory for output of all the steps. Outputs that are long are not included in this report.

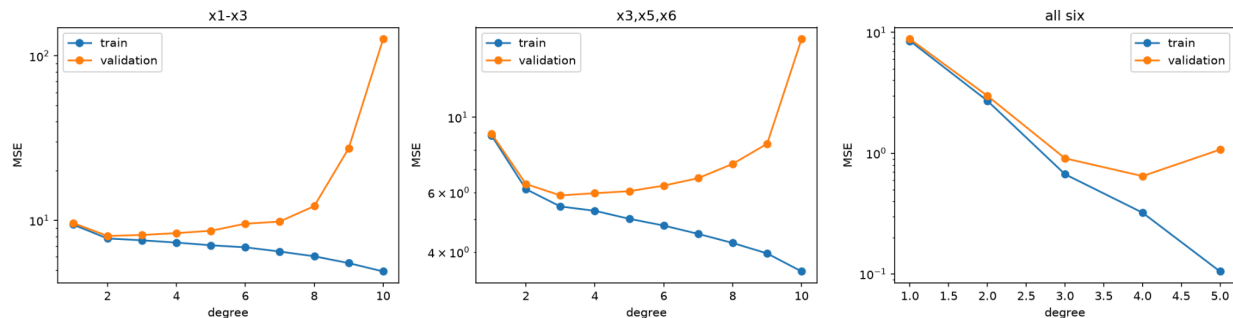
Phase 1

1. Exploratory Data Analysis (EDA)

Use 5-fold cross validation for different degrees and different sets of features.

Run **var1_eda.py**, **var1_sweep.py**

- The interior looks similar in both sets, but test has bigger spikes at ± 1 .
- All six features is by far the best. Validation MSE is 0.646 at degree 4, and 0.907 at degree 3.



2. Unregularized analysis

Check the best degree without any regularization using all 6 features.

Run **var1_unreg.py**

degree	n_terms	train_mse	val_mse	val_std	val_r2
1	6	8.51628	8.81640	0.63425	0.09399
4	3			1	6

2	27	2.71446 5	2.98329 2	0.51548 1	0.69592 7
3	83	0.66934 9	0.90670 5	0.11407 2	0.90690 5
4	209	0.32152 8	0.6456	0.05969 6	0.93378 7
5	461	0.10475	1.07001 2	0.22992 6	0.88903 4
6	923	2.70E-27	89.3736	18.7512 7	-8.30492

Best unregularized degree = 4

3. Regularization

Run **var1_reg.py**, **var1_lasso_deg.py**, **var1_enets.py**

Running Ridge on degrees 3–6 and Lasso on degrees 3–5, since Lasso gets slow with many terms.

And pushing lasso to degrees 6 and 7.

Check errors using elastic m=nets for degree 4, 5, 6.

- Lasso beats Ridge by a real margin at degree 5. Validation MSE is 0.294 for Lasso against 0.461 for Ridge. The gap is about 5× the per-fold std, so it's not noise. Both are well below the unregularized best of 0.646.
- Validation MSE rises with degree after 5: 0.294 → 0.311 → 0.328 for degrees 5, 6 and 7. Together with the earlier 0.901 and 0.562 at degrees 3 and 4, the curve goes down, bottoms out at 5, and goes back up.
- Degrees 5 and 6 are effectively tied.
- ElasticNet doesn't beat Lasso.
- The l1_ratio = 1.0 rows reproduce the Lasso numbers at degrees 5 and 6 (0.2940 and 0.3107)

4. Prediction

Final model: Lasso, degree 5, alpha = 0.01, all six features.

Run **var1_final.py**

- Final prediction saved as **IMT2024066_pred_var1.csv**

Phase 2

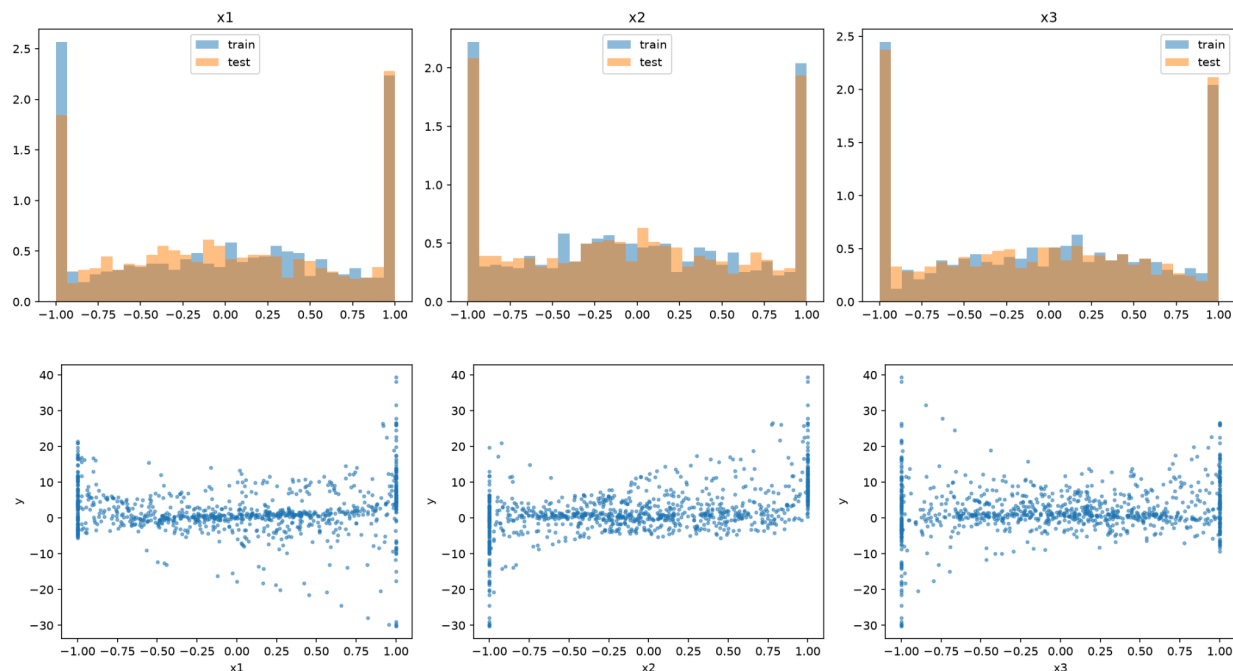
1. EDA + unregularized analysis

Use 5-fold cross validation for different degrees and different set of features.

Check the errors for all degrees and feature combinations without any regularization.

Run `var2_eda.py`, `var2_sweep.py`

- Only the full 3-feature model works. With x_1 , x_2 and x_3 together, validation MSE bottoms out at 0.295 (degree 8, R^2 0.994). The best any other subset manages is 18.9 (x_1+x_2).
- Degrees 7 to 9 are close. Degree 8 is 0.295 with a fold std of 0.031, degree 9 is 0.33 and degree 7 is 0.34.



2. Regularization

Run `var2_reg.py`, `var2_enets.py`

- The gain from regularizing is small. The best regularized score is 0.2787 (Ridge, degree 10, α 1), against 0.295 for the unregularized best (degree 8).
- Degrees 8 to 14 are all about equally good once regularized. The best-per-degree Ridge scores are 0.291, 0.279, 0.281 and 0.292 for degrees 8, 10, 12 and 14, all within one std of each other.
- Ridge and Lasso are about equal, with Ridge slightly ahead.
- The best ElasticNet score is 0.2783, against Ridge's 0.2787. That's a gap of 0.0004 with a fold std of about 0.036, which is a tie. By the general rule, a tie goes to the simpler model, which is Ridge.

4. Prediction

The final candidates are all Ridge:

- A: degree 8, alpha 0.0316 (CV 0.291)
- B: degree 10, alpha 1 (CV 0.279)
- C: degree 12, alpha 1 (CV 0.281)

Take **model B** since it has the least Validation error.

Run **var2_final.py**

- Final prediction saved as **IMT2024066_pred_var2.csv**